<DATA ANALYTICS WITH POWER BI >

PROJECT REPORT

(Project Semester: August- December 2025)

*(*Farm Productivity & Crop Performance Dashboard*)*

Submitted by

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Registration No 12320564

Section K23CP

Course Code INT374

Under the Guidance of

(Sadaf Qureshi)

Discipline of CSE/IT

Lovely School of Computer Science & Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Nippon Tadrishi bearing Registration no.12320564 has completed .INT374 project titled, **"Super Store sales Dashboard"** under my guidance and supervision of Sadaf Qureshi. To the best of my knowledge, the present work is the result of his original development, effort and study.

Name of the Supervisor – Sadaf Qureshi
Designation of the Supervisor- Assistant Professor
School of Computer science & Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 10 December 2025

DECLARATION

I, Nippon Tadrishi, student of Computer science & engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 10 December 2025 Name of the student

Registration No. 12320564 Nippon Tadrishi

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor and guide Sadaf Qureshi for their valuable guidance and support throughout this project. I am also thankful to Lovely Professional University for providing me with the platform and resources needed to complete this minor project. Lastly, I thank my peers and family for their encouragement.

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1. Introduction

In today's data-driven business environment, organizations rely heavily on data analytics to understand sales performance, identify trends, and support strategic decision-making. Retail businesses, in particular, generate large volumes of transactional data that can be transformed into valuable insights using effective visualization and forecasting techniques.

This project focuses on the analysis of **Superstore Sales data** through the development of an interactive dashboard. The dashboard provides a comprehensive view of sales performance across different regions, product categories, and time periods. Visual elements such as **maps and bar charts** are used to highlight geographical sales distribution and category-wise performance, while **forecasting techniques** are applied to predict future sales trends.

The primary objective of this report is to present how data visualization and analytical methods can convert raw sales data into meaningful insights. The dashboard helps stakeholders monitor key performance indicators (KPIs), identify high-performing and underperforming areas, and support data-driven business decisions. This project demonstrates the practical application of business intelligence tools in retail sales analysis.

2. Source of Dataset

The dataset used in this project was obtained from Kaggle, an open data platform widely used for data science and analytics projects.It includes key attributes such as device catagori, region, sub catagoriesd, state,and year.

The dataset covers major crops across different states and regions of India, providing a comprehensive view of agricultural performance and productivity trends. This structured dataset serves as the foundation for developing an interactive Power BI dashboard that enables analysis of crop performance, regional variations, and long-term agricultural trends.

3. Dataset Preprocessing

 Data pre-processing was performed to ensure accuracy and consistency before analysis and dashboard creation. The raw Superstore dataset was first inspected for **missing, duplicate, and inconsistent values**, which were cleaned or removed where necessary. Date fields were converted into a proper **date-time format** to enable time-based analysis and forecasting. Categorical variables such as region, category, and sub-category were standardized to maintain uniform naming. Numerical fields like sales, profit, and quantity were validated for outliers and data type correctness. After cleaning and transformation, the dataset was structured and optimized for effective visualization and analysis.

4. Analysis on Dataset

i. General Description:

The analysis was carried out using **Microsoft Power BI**, utilizing tools such as **Power Query, DAX measures, slicers, and interactive visualizations** to analyze Superstore sales data. The dashboard was designed to explore **sales and profit patterns, regional performance, product category-wise analysis, and time-based trends**, along with **sales forecasting** across different regions, segments, and time periods..

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ii. Specific Requirements and Analysis Results:

1. Overview Dashboard

The Super Store Sales Dashboard presents an interactive analysis of sales, profit, quantity, and delivery performance across regions, states, categories, and time periods. It uses maps, bar charts, line graphs, and KPIs to visualize regional trends, product performance, and time-based sales patterns. Interactive filters enable quick insights and support data-driven business decisions.

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2. Trend and Performance Analysis

- **Bar and column charts** are used to compare sales and profit across regions, states, categories, and sub-categories.
- **Line charts** illustrate month-wise and year-wise trends in sales and profit, helping identify growth patterns and seasonality.
- **Donut charts** display the distribution of sales by segment, ship mode, and payment mode.
- **Time-series and comparative charts** analyze order-date-wise sales performance to evaluate consistency and peak periods.
- **Interactive slicers** for Region, State, Category, Sub-Category, and Time Period enable dynamic and focused analysis.

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iii. Visualization:

All visualizations were developed using **Power BI's built-in visual tools**, including **bar charts, line charts, donut charts, tables, maps, and KPI cards**. Consistent formatting and layout design were applied to enhance readability, while **slicers and filters** improve interactivity and allow users to drill down into specific **regions, categories, and time periods**. This visual approach ensures that complex sales data is presented in a **clear, intuitive, and insight-driven** manner.

5. Conclusion

This project demonstrated how **Microsoft Power BI** can be effectively used for the analysis and visualization of **retail sales data**. By leveraging **Power Query** for data preprocessing, **DAX** for calculations, and interactive visuals such as **charts, maps, tables, KPI cards, and slicers**, a comprehensive **Super Store Sales Dashboard** was developed. The dashboard provides clear insights into **sales and profit trends, product category performance, regional distribution, and time-based patterns**, along with forecasting support. This project highlights the practical application of **business intelligence tools in retail analytics** and emphasizes how data-driven insights can support informed decision-making and performance evaluation.

6. Future Scope

Future enhancements of this project could include integrating **real-time sales and inventory data sources** into Power BI to enable continuous monitoring of business performance. Advanced analytics techniques such as **Python-based analysis and machine learning models** can be applied to **forecast sales demand, predict customer purchasing behavior, and optimize inventory levels**. Additionally, expanding the dataset to include **customer demographics, promotional campaigns, supplier data, and shipping performance metrics** could further improve the depth of analysis and support **data-driven planning and strategic decision-making** in the retail domain.

7. References

[1] Kaggle, **"Superstore Sales,"** Kaggle Dataset. The dataset was accessed for academic and analytical purposes and used to analyze products sales, profit region, sales trends, and regional performance across United states. Available: https://www.kaggle.com/datasets/vivek468/superstore-dataset-final

[2] Microsoft Support, **"Power BI Documentation,"** [Online]. Available: https://community.tableau.com/s/question/0D54T00000CWeX8SAL/sample-superstore-sales-excelxls

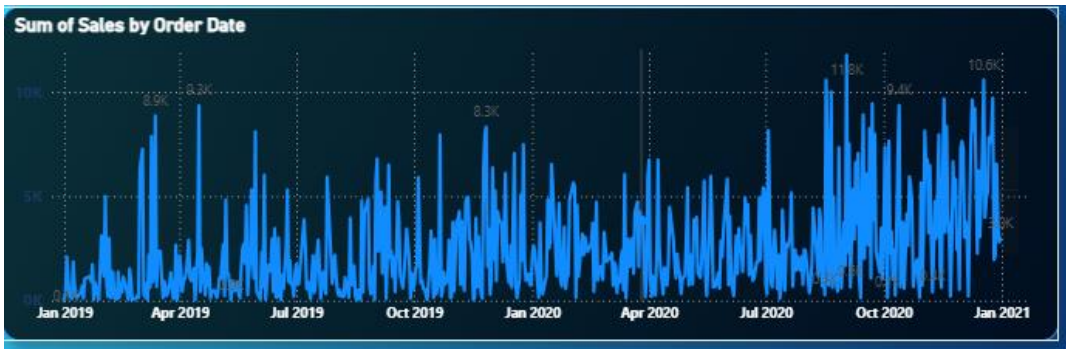
[3] Microsoft Support, **"Data Analysis Expressions (DAX) Reference,"** [Online]. Available: https://learn.microsoft.com/en-us/dax/

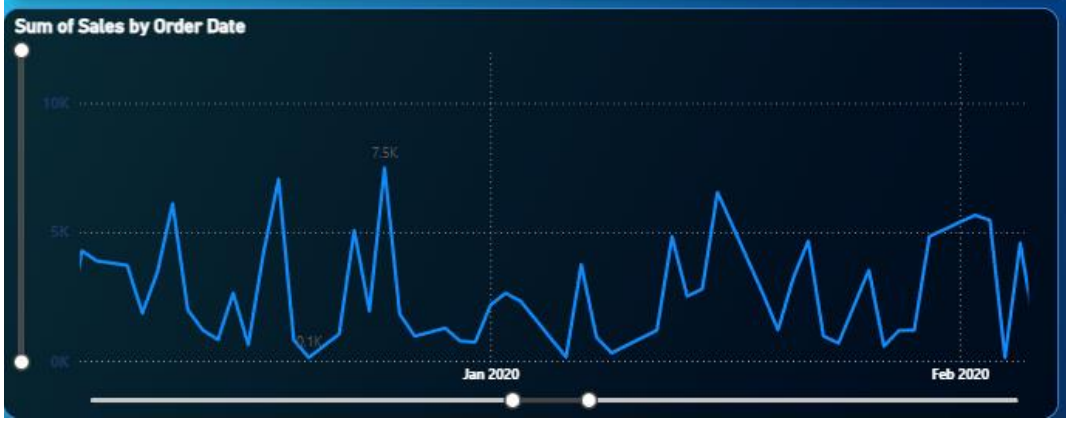
8. LinkedIn post - https://www.linkedin.com/posts/nippon-tadrishi_dataanalytics-powerbi-dashboarddesign-activity-7406379389766098945-1QyV?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEYbq5EBzE0taAIEMMV6dTGysJ4PMH_WeMM

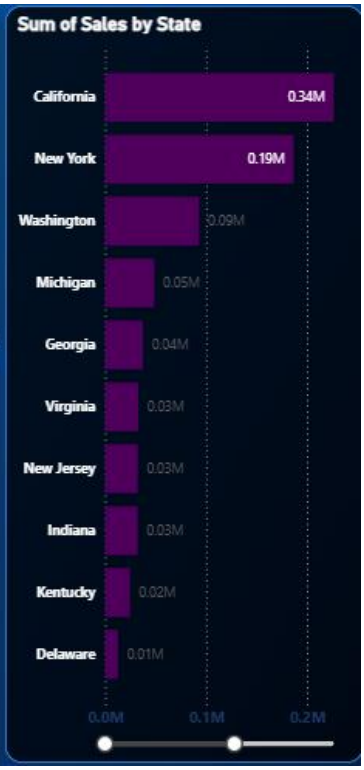
5. Dataset URL- https://www.kaggle.com/datasets/vivek468/superstore-dataset-final

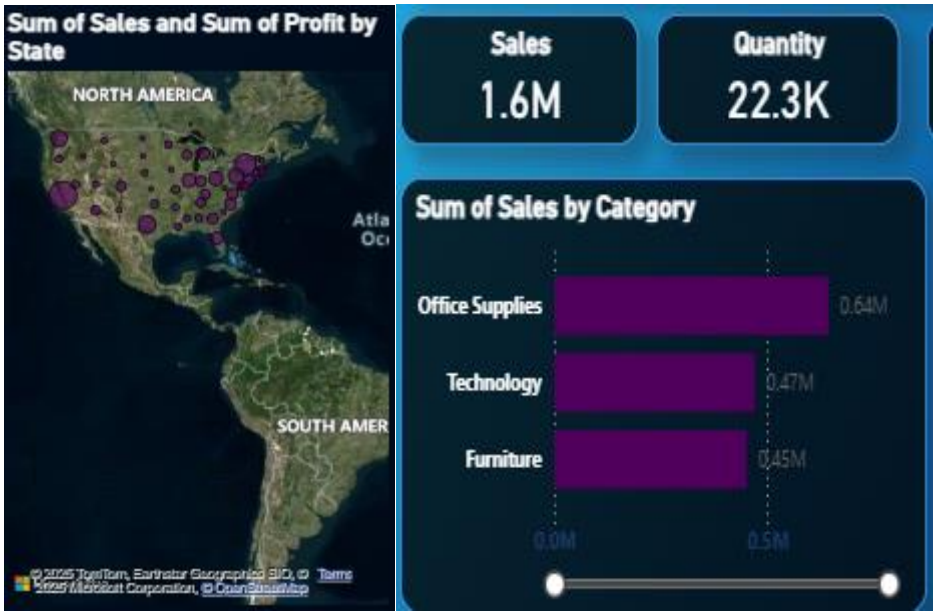
Dashboad

Super Store sales dashboard
Select all
Central
East
South
West
Sum of Sales and Sum of Profit by State
NORTH AMERICA
SOUTH AMER
Sum of Sales by Ship Mode
81.66K (14.75%)
111.67K (20.1...)
329... (59...)
Sum of Sales by Payment Mode
344.3... (21...)
667.42K (42.62%)
553.99K (35.38%)
Sum of Sales by Segment
303.06K (19.35%)
753K (48.0...)
509.74K (32.55%)
Sales
1.6M
Quantity
22.3K
Profit
175.3K
Avg Delivery Days
3.9
Sum of Sales by Category
Office Supplies
0.64M
Technology
0.47M
Furniture
0.45M
Sum of Sales by Sub-Category
Phones
0.20M
Chairs
0.18M
Binders
0.17M
Storage
0.15M
Accessories
0.12M
Sum of Profit by Month and Year
Sum of Sales by Month and Year
January
February
March
April
May
June
July
August
September
October
November
December


Sum of Sales by Order Date
Jan 2019
Apr 2019
Jul 2019
Oct 2019
Jan 2020
Apr 2020
Jul 2020
Oct 2020
Jan 2021


Sum of Sales by Order Date
7.5K
Jan 2020
Feb 2020


Sum of Sales by State
California
0.34M
New York
0.19M
Washington
0.09M
Michigan
0.05M
Georgia
0.04M
Virginia
0.03M
New Jersey
0.03M
Indiana
0.03M
Kentucky
0.02M
Delaware
0.01M
0.0M
0.1M
0.2M


Sum of Sales and Sum of Profit by State
NORTH AMERICA
SOUTH AMER
Sales
1.6M
Quantity
22.3K
Sum of Sales by Category
Office Supplies
0.64M
Technology
0.47M
Furniture
0.45M
0.0M
0.5M